

Programming Assignment #4 (Due on day 7 of Module 9)

Binary Classification

This assignment has several parts and it is due in three weeks. The necessary data files are available in Canvas.

For this lab you will work with a set of multilingual movie reviews.¹ The task is focused on building and evaluating binary classifiers using the training data and to then use the trained model to make predictions about whether reviews express primarily positive or negative sentiment. There are no neutral reviews in the data that you will be working with. You have flexibility in choosing which methods you use for the classification task, but you will be asked to train a model, make predictions, and evaluate the accuracy of predictions. A key objective is for you to gain experience using supervised machine learning tools for text classification tasks. Unlike previous programs, you are expected to use existing tools instead of implementing your own system from scratch.

Data

The datasets are available as JSONL files. Each line of the file consists of three fields described below. The data are split into two portions: **train** and **test**. There are separate files for English reviews and for Spanish reviews. The train portion should be used to build your models. The test partition is used to evaluate your models. The splits are roughly balanced between positive and negative examples.

Field	Description
docid	A unique hexadecimal ID for the review.
text	UTF8 encoded text of a movie review. Depending on the source, the text might be in English or in Spanish.
label	Either 0, indicating negative sentiment, or 1, indicating positive sentiment.

Filename	Rows
eng.imdb.train.jsonl	20,000
eng.imdb.test.jsonl	1,000
spa.muchocine.train.jsonl	1,625
spa.muchocine.test.jsonl	1,000

Example lines (English):

```
{
  "id": "d8bc9240e79d3db09bcb0f9994644980",
  "text": "Being a long-time Steve Martin fan, it hurt to see him in a movie with such a cliched script. The screenwriter must have just rented some videos and taken a piece here and a piece there - it certainly is about the least imaginative movie I've seen in a long time - I knew exactly what was going to happen in each scene as soon as it began. The African-American stereotypes and slave references are pretty offensive as well. But the thing that was the worst for me was to see Martin in a movie without a trace of wit. He probably is one of the most intelligent actors and here he totally sold out to a totally dumbed down script. He must be behind in his alimony to be in such a lame effort.",
  "label": 0
}
{
  "id": "37ac486345e7aeb4ec24c414d2470e",
  "text": "A masterpiece. Thus it is, possibly, not for everyone. The camera work, acting, directing and everything else is unique, original, superb in every way - and very different from the trash we are sadly used to getting. Summer Phoenix creates a deep, believable and intriguing Esther Kahn. As everything else in this film, her acting is unique - it is completely her own - neither \"British\" nor \"American\" nor anything else I have ever seen. There is something mesmerizing about it. The lengthy, unbroken, natural shots are wonderful, reminding us that we have become too accustomed to a few restricted ways of shooting and editing.",
  "label": 1
}
```

You can use any classification method (e.g., Naïve Bayes, decision trees, SVMs, regular expressions, neural networks, KNN, etc...). You are strongly encouraged to work with open source machine learning toolkits such as NLTK, sklearn, spaCy, SVMLight, HuggingFace, etc.... Here are a couple of pointers:

- <https://pythonprogramming.net/text-classification-nltk-tutorial/>
- <https://www.nltk.org/book/ch06.html> (through section 1.3)
- <https://pythonprogramming.net/sentiment-analysis-module-nltk-tutorial/>
- <https://www.dataquest.io/blog/tutorial-text-classification-in-python-using-spacy/>

¹ This is real user-generated content. There could be vulgar language or objectionable content.

Tasks

(a) Study the English training data (20 points)

- Identify some features that are likely to be useful indicators of either positive or negative sentiment. Give 10 example features and the relative frequency seen in both classes (e.g., wonderful, 16.4% (pos), 3.6% (neg).)
- Report any other observations about the data that are helpful for this task.

(b) Train an English classifier and make predictions (20 points)

Using the training partition build a supervised model using a learning model of your choice (e.g., Naive Bayes, SVMs, KNN, neural networks, etc...). Then make predictions on the English test partition.

- Succinctly describe your approach to tokenization and featurization. Include details such as the tool you used, the algorithm employed, important parameters, the type of features, and the total number of features.
- Print *docid [tab] prediction* for only the first 10 documents in the test file. Prediction should be 0 or 1.

(c) Evaluate your predictions (20 points)

For predictions using your model in part (b) compute precision, recall and F_1 scores for both the 0 (negative) and 1 (positive) classes over the entire English test partition. Show the work in your computation (*i.e.*, the numerators and denominators for precision and recall).

- Calculate and report Precision, Recall, and F_1 scores.
- Find a couple of mistakes that the classifier makes. Present a couple of incorrect predictions that you find interesting along with a short comment.

(d) Train and evaluate a Spanish classifier (20 points)

This is the same procedure that you did in steps (b) and (c) above. Here the only thing that is changing is the language of the reviews. Note the Spanish training data is smaller than the English training set. Report the same information that was requested for the English data:

- Succinctly describe your approach to tokenization and featurization. Include details such as the tool you used, the algorithm employed, important parameters, the type of features, and the total number of features.
- Print *docid [tab] prediction* for the first 10 documents only from the Spanish test set.
- As before calculate and report Precision, Recall, and F_1 scores.

(e) Cross-language classification (20 points)

Perform classification on the Spanish test set *without* training using the Spanish training reviews. Below are some alternative methods that you may choose from. You only need to perform one approach. Report the same information requested in (d).

- Use only the English training data with character n-grams (e.g., 4-grams) as features instead of words. The Spanish text will contain some matches for English n-grams (e.g., *perfecta* is a translation of perfect; *repulsivo* is a translation of repulsive).
- Take a set of English sentiment words and translate them into Spanish (somehow). Use the English training data, but modify it so that these terms are translated to Spanish (e.g., by dictionary lookup). These translated Spanish terms can be your features. I am providing an English resource, SentiWordNet that might be helpful.
- Translate the Spanish test reviews into English using Machine Translation, and use your previously trained English sentiment model on the translated reviews. This would be a bit complex and would involve using an open source MT model or API (e.g., translate-shell, EasyNMT, OpenNMT, NLLB).
- Find a set of Spanish sentiment terms and use simple logic to make classification decisions. For example, given say 20 common positive terms and 20 common negative terms, pick whichever class has the most terms present in the review.
- Any other technique that doesn't use the Spanish training data but does classify the Spanish test reviews.

You should submit a single PDF file in Canvas that contains the requested output for the tasks above, as well as any code that you write. Be sure to cite any resources or toolkits that you make use of.